

MH3100 Real Analysis I

Midterm Practice 4 (from past Finals) (Solutions)

Past Final questions in Midterm scope (Lectures 1–17)

Quantitative Research Society @NTU

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About This Practice Paper

This is QRS@NTU's fourth MH3100 Midterm practice paper. Instead of a fresh QRS-authored paper, it collects every question part from NTU's three most recent MH3100 Real Analysis I *Final* examinations (AY2022/23, AY2023/24, AY2024/25, Semester 2) that falls inside the **Midterm scope, Lectures 1–17** (construction and completeness of $\mathbb{R}$ through sets of real numbers). Parts that require Lecture 18–20 material (function limits, continuity, uniform continuity), differentiation, integration, uniform convergence of function sequences/series, or open covers/Heine-Borel are outside the Midterm syllabus and are omitted below; the scope table gives the reason for every omission. Since this paper draws only on genuine past-Final material, its total (97 marks across 8 questions) does not match the 24-mark, 60-minute Midterm format – treat it as extra practice, not a timed mock.

Solutions in this paper are held to the same three writing expectations as the Midterm: complete sentences; related statements joined by logical connectives; and inference explicitly marked with "therefore/hence/so".

Overview

Papers used (all MH3100 Real Analysis I, NTU, CLOSED BOOK, 6 questions, 2 hours):

- Semester 2 Examination 2022–2023 (April 2023).
- Semester II Examination 2023–2024 (April 2024).
- Semester II Examination 2024–2025 (April 2025).

In every one of the three papers, Question 1 (15 marks, five True/False parts) and Question 2 (15 marks, three existential parts) mix in-scope and out-of-scope sub-parts, Question 3 (20 marks) is a supremum/infimum proof that is fully in scope, and Questions 4–6 (20+20+10 marks) all hinge on continuity and/or uniform convergence and are fully out of scope. Marks are not printed per sub-part on the original papers for Questions 1–2; QRS distributes each question's total evenly across its parts.

Scope table

AY2022/23 Semester 2 Final

Question	Status	Reason
Q1(a)–(c)	Partly (3/5)	Monotone Convergence Lemma (L9); Theorem on Subsequential Limits (L7); limit points (L13) – all in scope.
Q1(d)–(e)	Out	Uniform continuity; power series and uniform convergence – beyond L17.
Q2(a)	Partly (1/3)	Absolute convergence + bounded sequence, via Comparison/Absolute Convergence Test (L12).
Q2(b)–(c)	Out	Function limits (L18–20 topic); pointwise convergence of continuous functions – beyond L17.
Q3	In	Definition of supremum (L2), fully in scope.
Q4	Out	Hypothesis “ f continuous” is L18–20 material.
Q5	Out	Uniform convergence of a function series – not covered in L1–17.
Q6	Out	Uniform convergence of continuous functions – not covered in L1–17.

AY2023/24 Semester 2 Final

Question	Status	Reason
Q1(a)–(c)	Partly (3/5)	Suprema of sets (L2); Bolzano-Weierstrass Theorem (L8); isolated points/closed sets (L13, L15) – all in scope.
Q1(d)–(e)	Out	Differentiability; uniform convergence of derivatives – beyond L17.
Q2(a)	Partly (1/3)	Series with $\sum a_n$ convergent but $\sum a_n^2$ not (L10–12).
Q2(b)–(c)	Out	Continuity and uniform continuity – beyond L17.
Q3	In	Definition of supremum (L2), fully in scope.
Q4	Out	Continuous functions f, g – L18–20 material.
Q5	Out	Uniform convergence of a function series.
Q6	Out	Function limits (L18–20 topic).

AY2024/25 Semester 2 Final

Question	Status	Reason
Q1(a)–(c)	Partly (3/5)	Sup/inf (L2); Cauchy Criterion for Series (L10) with the Archimedean Property (L3); closure/closed sets (L14–15) – all in scope.
Q1(d)–(e)	Out	Function limits (L18–20 topic); Weierstrass- M -test-style uniform convergence.
Q2	Out (0/3)	(a)–(b) continuous-function hypotheses; (c) uniform convergence – all beyond L17.
Q3	In	Sup/inf via rational upper bounds (L2) + Archimedean Property (L3), fully in scope.
Q4	Out	Continuous functions and compactness combined; the continuity hypothesis is L18–20 material (compactness itself, L16, would be in scope).
Q5	Out	Uniform convergence of a function series.
Q6	Out	Continuity + uniform convergence.

Total marks in this practice paper

Question	1	2	3	4	5	6	7	8	Total
Marks	9	5	20	9	5	20	9	20	97

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Question 1

(9 Marks)

For each of the following statements, determine if it is true or false. You do not need to justify your answer.

- (a) A monotone sequence is convergent if and only if it has a convergent subsequence.
- (b) A sequence is divergent if it has a divergent subsequence.
- (c) If a set has no limit points, then it is finite.

(Source: MH3100 Final AY22-23 Sem 2, Question 1(a)–(c). Question 1(d)–(e), on uniformly continuous functions and uniform convergence of power series, fall outside the Lecture 1–17 Midterm scope and are omitted; see the scope table above. The Final prints only the question total, 15 marks, for all five parts; QRS distributes this evenly, 3 marks per part.)

Solution

- (a) ☐ True

Proof. $(\Rightarrow)$ $\{a_n\}$ converges $\therefore \{a_n\}$ is a convergent subsequence of itself ($n_k = k$). $(\Leftarrow)$ $\{a_n\}$ is monotone with a convergent subsequence $\therefore \{a_n\}$ converges (Monotone Convergence Lemma). $\square$

- (b) ☐ True

Proof. Let $\{a_{n_k}\}$ be a divergent subsequence of $\{a_n\}$, and suppose $\{a_n\}$ converges, with limit L . $\therefore L = \lim a_{n_k}$ (Subsequential Limits), contradicting divergence of $\{a_{n_k}\}$. $\therefore \{a_n\}$ diverges. $\square$

- (c) ☐ False

Proof. $F := \mathbb{Z}$ is infinite. Suppose $a \in \mathbb{R}$ is a limit point of $\mathbb{Z}$. $\therefore$ some $x \in \mathbb{Z} \setminus \{a\}$ has $|x - a| < \frac{1}{2}$. $\therefore \varepsilon := |x - a| > 0$, and some $y \in \mathbb{Z} \setminus \{a\}$ has $|y - a| < \varepsilon$. $\therefore y \neq x$ and $|x - y| \leq |x - a| + |a - y| < \frac{1}{2} + \frac{1}{2} = 1$, contradicting that distinct integers differ by at least 1. $\therefore \mathbb{Z}$ has no limit points, yet $\mathbb{Z}$ is infinite. $\square$

Marking scheme (9 marks)

QRS suggested marking scheme.

- (a) **(3 marks)** 3 marks for ☐ True; 0 marks for ☐ False. No justification is required on the actual exam.
- (b) **(3 marks)** 3 marks for ☐ True; 0 marks for ☐ False.
- (c) **(3 marks)** 3 marks for ☐ False; 0 marks for ☐ True.

Typical deductions (if justification is examined): in (a), proving only one direction of the "if and only if"; in (b), attempting to prove convergence directly instead of arguing by contradiction via the Theorem on Subsequential Limits; in (c), searching only among finite sets instead of recognising that infinite discrete sets such as $\mathbb{Z}$ or $\mathbb{N}$ already settle the claim.

Question 2

(5 Marks)

For each of the following existential statements, either give an example fulfilling the statement or prove that it is not true.

There is an absolutely convergent series $\sum_{n=1}^{\infty} a_n$ and bounded sequence $\{b_n\}_{n=1}^{\infty}$ for which the series $\sum_{n=1}^{\infty} a_n b_n$ is divergent.

(Source: MH3100 Final AY22-23 Sem 2, Question 2(a). Question 2(b)–(c), on function limits and pointwise convergence of continuous functions, fall outside the Lecture 1–17 Midterm scope and are omitted; see the scope table above. The Final's Question 2 total, 15 marks, is distributed evenly, 5 marks per part.)

Solution

Proof. The statement is false: let $\sum |a_n|$ converge and $\{b_n\}$ be bounded, so $|b_n| \leq M$ for all $n \in \mathbb{N}$, some $M > 0$.

$$0 \leq |a_n b_n| = |a_n| |b_n| \leq M |a_n| \quad \text{for all } n \in \mathbb{N}$$

$\sum M |a_n|$ converges (ALT) $\therefore \sum |a_n b_n|$ converges (Comparison Test) $\therefore \sum a_n b_n$ converges (Absolute Convergence Test). So no such series and sequence exist. $\square$

Marking scheme (5 marks)

QRS suggested marking scheme.

- **1 mark:** correctly recognising the statement is false, not attempting to fabricate an example.
- **1 mark:** introducing the bound $M > 0$ with $|b_n| \leq M$ and the pointwise estimate $|a_n b_n| \leq M |a_n|$.
- **1 mark:** correct use of the Comparison Test to conclude $\sum |a_n b_n|$ converges.
- **1 mark:** correct use of the Absolute Convergence Test to conclude $\sum a_n b_n$ converges.
- **1 mark:** the three writing expectations are met throughout.

Typical deductions: citing the Comparison Test with the inequality reversed; concluding convergence of $\sum a_n b_n$ without the intermediate absolute-convergence step; using a non-bounded or only conditionally convergent example.

Question 3

(20 Marks)

Use the definition of supremum to prove that if two nonempty sets $A, B \subseteq \mathbb{R}$ that are bounded above satisfy $\sup A < \sup B$, then for all $a \in A$, there exists $b \in B$ such that $a < b$. Give an example to show that the converse is not true.

You do not need to show that the suprema exist.

(Source: MH3100 Final AY22-23 Sem 2, Question 3, verbatim, 20 marks.)

Solution

Method 1: definition of supremum

Proof. Let $s := \sup A < t := \sup B$ and $a \in A$; $a \leq s$ (s upper bound of A). Suppose no $b \in B$ has $a < b$ $\therefore b \leq a$ for all $b \in B$ $\therefore a$ is an upper bound of B $\therefore t \leq a$ (t least upper bound).

$$t \leq a \leq s < t \quad (\text{contradiction})$$

$\therefore$ some $b \in B$ has $a < b$; $a \in A$ was arbitrary. □

Converse fails. $A = B := (0, 1)$. For $a \in A$: $b := \frac{a+1}{2}$ has $a < b < 1$ $\therefore b \in B$, $a < b$. But $\sup A = 1 = \sup B$ $\therefore \sup A < \sup B$ fails.

Method 2: epsilon-Criterion for Suprema

Let $a \in A$ $\therefore a \leq s < t \implies \varepsilon := t - a > 0 \implies$ some $b \in B$ has $b > t - \varepsilon = a$ (ε -Criterion for $t = \sup B$). $a \in A$ was arbitrary.

Marking scheme (20 marks)

QRS suggested marking scheme.

- **4 marks:** correct definition of supremum (upper bound; not greater than any upper bound) applied to both $s = \sup A$ and $t = \sup B$.
- **4 marks:** setting up the proof by contradiction, assuming a is an upper bound for B .
- **4 marks:** correctly deriving $t \leq a$ from the minimality of $t = \sup B$ among upper bounds of B .
- **4 marks:** correctly combining $t \leq a \leq s < t$ to reach the contradiction, and concluding the existence of $b \in B$ with $a < b$.
- **3 marks:** a valid example (with justification) showing the converse fails.
- **1 mark:** the three writing expectations are met throughout.

Typical deductions: proving only that a is not an upper bound for B without extracting the required $b \in B$ with $a < b$; choosing a converse example in which $\sup A < \sup B$ still holds.

Question 4

(9 Marks)

For each of the following statements, determine if it is true or false. You do not need to justify your answer.

- (a) If the set A has supremum s and the set B has supremum t , then their intersection $A \cap B$ has supremum $\min\{s, t\}$.
- (b) If an unbounded sequence has a bounded subsequence, then it has a convergent subsequence.
- (c) If a set contains only isolated points, then it is closed.

(Source: MH3100 Final AY23-24 Sem 2, Question 1(a)–(c). Question 1(d)–(e), on differentiability and uniform convergence of derivatives, fall outside the Lecture 1–17 Midterm scope and are omitted; see the scope table above. Marks are again distributed evenly, 3 marks per part.)

Solution

- (a) False

Proof. $A := (0, 1)$, $B := (2, 3)$: $\sup A = 1$, $\sup B = 3 \therefore \min\{\sup A, \sup B\} = 1$. $A \cap B = \emptyset$: every $r \in \mathbb{R}$ is an upper bound and $r - 1 < r$ is a smaller one $\therefore \emptyset$ has no least upper bound. $\therefore A \cap B$ has no supremum, so none equal to 1. $\square$

- (b) True

Proof. Let $\{a_{n_k}\}$ be a bounded subsequence $\therefore$ it has a convergent subsequence $\{a_{n_{k_j}}\}$ (Bolzano–Weierstrass). $n_{k_1} < n_{k_2} < \dots \therefore \{a_{n_{k_j}}\}$ is a subsequence of $\{a_n\}$. $\square$

- (c) False

Proof. $F := \{\frac{1}{n} : n \in \mathbb{N}\}$. Isolated: let $n \in \mathbb{N}$, $a := \frac{1}{n}$, $\varepsilon := \frac{1}{n(n+1)}$, and $m \in \mathbb{N}$, $m \neq n$.

$$\begin{aligned} m > n &\implies a - \frac{1}{m} \geq \frac{1}{n} - \frac{1}{n+1} = \varepsilon & \left(\frac{1}{m} \leq \frac{1}{n+1}\right) \\ m < n &\implies \frac{1}{m} - a \geq \frac{1}{n-1} - \frac{1}{n} = \frac{1}{n(n-1)} > \varepsilon & \left(\frac{1}{m} \geq \frac{1}{n-1}\right) \end{aligned}$$

$\therefore$ no $x \in F \setminus \{a\}$ has $|x - a| < \varepsilon \therefore$ every point of F is isolated. Let $\varepsilon > 0$ and $N > \frac{1}{\varepsilon}$ (Archimedean) $\therefore \frac{1}{N} \in F \setminus \{0\}$ and $|\frac{1}{N} - 0| < \varepsilon \therefore 0$ is a limit point of F . $0 \notin F \therefore F$ is not closed. $\square$

Marking scheme (9 marks)

QRS suggested marking scheme.

- (a) **(3 marks)** 3 marks for False; 0 marks for True.
- (b) **(3 marks)** 3 marks for True; 0 marks for False.

(c) **(3 marks)** 3 marks for ☐ False; 0 marks for ☐ True.

Typical deductions: in (a), forgetting that an empty intersection has no real supremum; in (b), forgetting that a subsequence of a subsequence is still a subsequence of the original sequence; in (c), confusing "every point is isolated within the set" with "the set has no limit points at all outside itself".

Question 5

(5 Marks)

For each of the following existential statements, either give an example fulfilling the statement or prove that it is not true.

There is a convergent series $\sum_{n=1}^{\infty} a_n$ such that the series $\sum_{n=1}^{\infty} a_n^2$ is not convergent.

(Source: MH3100 Final AY23-24 Sem 2, Question 2(a). Question 2(b)–(c), on continuous functions and Cauchy sequences and on uniform continuity, fall outside the Lecture 1–17 Midterm scope and are omitted; see the scope table above. Marks distributed evenly, 5 marks per part.)

Solution

Proof. The statement is true: take $a_n := \frac{(-1)^n}{\sqrt{n}}$. $\frac{1}{\sqrt{n}}$ is decreasing ($\sqrt{n}$ increasing). Let $\varepsilon > 0$, $N > \frac{1}{\varepsilon^2}$ (Archimedean): for $n \geq N$, $\frac{1}{\sqrt{n}} \leq \frac{1}{\sqrt{N}} < \varepsilon \therefore 0 = \lim \frac{1}{\sqrt{n}} \therefore \sum a_n$ converges (Alternating Series Test). $a_n^2 = \frac{1}{n}$; let $s_m := \sum_{k=1}^m \frac{1}{k}$ for $m \in \mathbb{N}$:

$$s_{2m} - s_m = \frac{1}{m+1} + \cdots + \frac{1}{2m} \geq m \cdot \frac{1}{2m} = \frac{1}{2} \quad (\text{each term} \geq \frac{1}{2m})$$

$\therefore$ for $\varepsilon = \frac{1}{2}$ and every $N \in \mathbb{N}$, $n = N$, $m = 2N$ give $|s_m - s_n| \geq \varepsilon \therefore \sum a_n^2$ fails the Cauchy Criterion for Series $\therefore \sum a_n^2$ diverges. $\square$

Marking scheme (5 marks)

QRS suggested marking scheme.

- **1 mark:** a valid example built on conditional-but-not-absolute convergence.
- **2 marks:** correct proof that $\sum a_n$ converges (Alternating Series Test, with the hypothesis that the terms decrease to 0 verified).
- **1 mark:** correct proof that $\sum a_n^2 = \sum 1/n$ diverges (Cauchy Criterion for Series, tail estimate).
- **1 mark:** the three writing expectations are met throughout.

Typical deductions: an example where both series behave the same way; omitting the decreasing-terms check before invoking the Alternating Series Test.

Question 6

(20 Marks)

Use the definition of supremum to prove that if a set $A \subseteq \mathbb{R}$ is nonempty and bounded above and B denotes the set $\{b \in \mathbb{R} : \exists a \in A, b < a\}$, then B has a supremum $\sup B$ and the supremum satisfies $\sup B = \sup A$.

(Source: MH3100 Final AY23-24 Sem 2, Question 3, verbatim, 20 marks. Unlike Question 3 of the other two Finals in this paper, this question does not waive the requirement to show the supremum exists.)

Solution

Method 1: definition of supremum

Proof. $s := \sup A$ exists (AoC: A nonempty, bounded above).

Step 1: $B \neq \emptyset$. Take $a_0 \in A \therefore a_0 - 1 < a_0 \therefore a_0 - 1 \in B$.

Step 2: s is an upper bound of B . Let $b \in B \therefore$ some $a \in A$ has $b < a$; $a \leq s$ (s upper bound of A) $\therefore b < s$.

B is nonempty and bounded above $\therefore t := \sup B$ exists (AoC).

Step 3: $t \leq s$. s is an upper bound of B (Step 2) and t is the least one $\therefore t \leq s$.

Step 4: $s \leq t$. Suppose $t < s$. t is not an upper bound of A (else $s \leq t$, as s is least) $\therefore$ some $a \in A$ has $a > t$. $b^* := \frac{t+a}{2}$ has $t < b^* < a \therefore b^* \in B$ and $b^* > t$, contradicting that t is an upper bound of B . $\therefore s \leq t$.

Steps 3, 4 $\implies \sup B = t = s = \sup A$. □

Marking scheme (20 marks)

QRS suggested marking scheme.

- **3 marks:** Step 1 (B nonempty).
- **4 marks:** Step 2 (B bounded above by $s = \sup A$), with the definition of B correctly unpacked.
- **3 marks:** correctly invoking the Axiom of Completeness to obtain $t = \sup B$.
- **4 marks:** Step 3 ($t \leq s$).
- **5 marks:** Step 4 ($s \leq t$), including the proof-by-contradiction structure and the midpoint construction b^* .
- **1 mark:** the three writing expectations are met throughout.

Typical deductions: asserting $\sup B$ exists without first showing B is nonempty and bounded above; omitting the contrapositive step " $t < s \implies t$ is not an upper bound for A " in Step 4.

Question 7

(9 Marks)

For each of the following statements, determine if it is true or false. You do not need to justify your answer.

- (a) Two sets $A, B \subseteq \mathbb{R}$ that satisfy $(\forall a \in A, \forall b \in B, a < b)$ must satisfy $\sup A < \inf B$.
- (b) A series $\sum_{n=1}^{\infty} a_n$ that satisfies $(m > n \implies |a_{n+1} + a_{n+2} + \cdots + a_m| < \frac{1}{n})$ for all natural numbers $m, n \in \mathbb{N}$ is convergent.
- (c) A set is closed if and only if it is the closure of some set.

(Source: MH3100 Final AY24-25 Sem 2, Question 1(a)–(c). Question 1(d)–(e), on function limits and uniform convergence of a series of functions, fall outside the Lecture 1–17 Midterm scope and are omitted; see the scope table above. Marks distributed evenly, 3 marks per part.)

Solution

- (a) ☐ False

Proof. $A := (0, 1)$, $B := (1, 2)$: $a < 1 < b$ for all $a \in A$, $b \in B$ $\therefore$ the hypothesis holds. $\sup A = 1 = \inf B$ $\therefore$ $\sup A < \inf B$ reads $1 < 1$, false. $\square$

- (b) ☐ True

Proof. Let $\varepsilon > 0$; choose $N \in \mathbb{N}$ with $N > \frac{1}{\varepsilon}$ (Archimedean). For all $m > n \geq N$:

$$|a_{n+1} + \cdots + a_m| < \frac{1}{n} \leq \frac{1}{N} < \varepsilon \quad (\text{hypothesis; } n \geq N)$$

$\therefore \sum a_n$ converges (Cauchy Criterion for Series). $\square$

- (c) ☐ True

Proof. $(\Leftarrow) F = \overline{A} \implies \overline{F} = \overline{\overline{A}} = \overline{A} = F$ (Closure of a Closure) $\implies F \supseteq \overline{F} \therefore F$ is closed (Closure Criterion). $(\Rightarrow) F$ closed $\implies \overline{F} \subseteq F$ (Closure Criterion); $F \subseteq \overline{F}$ always $\therefore \overline{F} = F$, so F is the closure of F . $\square$

Marking scheme (9 marks)

QRS suggested marking scheme.

- (a) **(3 marks)** 3 marks for ☐ False; 0 marks for ☐ True.
- (b) **(3 marks)** 3 marks for ☐ True; 0 marks for ☐ False.
- (c) **(3 marks)** 3 marks for ☐ True; 0 marks for ☐ False.

Typical deductions: in (a), confusing strict inequality between every pair with strict inequality between the sup and inf; in (b), forgetting to choose N via the Archimedean Property so that $1/N < \varepsilon$; in (c), proving only one direction of the biconditional.

Question 8**(20 Marks)**

Use the definitions of suprema and infima to prove that if a set $A \subseteq \mathbb{R}$ and its set $B := \{b \in \mathbb{Q} : \forall a \in A, a \leq b\}$ of all rational upper bounds are nonempty, then $\sup A = \inf B$.

You do not need to show that the supremum and infimum exist.

(Source: MH3100 Final AY24-25 Sem 2, Question 3, verbatim, 20 marks.)

Solution**Method 1: definitions of supremum and infimum**

Lemma (a rational between two reals). If $x < y$ in $\mathbb{R}$, then $x < q < y$ for some $q \in \mathbb{Q}$.

Proof of Lemma. Choose $n \in \mathbb{N}$ with $\frac{1}{n} < y - x$ and $M \in \mathbb{N}$ with $M > |nx|$ (Archimedean) $\therefore nx + M > 0$. $S := \{k \in \mathbb{N} : k > nx + M\} \neq \emptyset$ (Archimedean) $\therefore S$ has a least element k_0 (Well-Ordering). If $k_0 \geq 2$: $k_0 - 1 \in \mathbb{N} \setminus S \implies k_0 - 1 \leq nx + M$; if $k_0 = 1$: $k_0 - 1 = 0 < nx + M$. $\therefore k_0 - 1 \leq nx + M < k_0$. $k := k_0 - 1 - M \in \mathbb{Z} \implies k \leq nx < k + 1$. Let $q := \frac{k+1}{n} \in \mathbb{Q}$.

$$nx < k + 1 \implies x < q$$

$$k \leq nx \implies q = \frac{k}{n} + \frac{1}{n} \leq x + \frac{1}{n} < x + (y - x) = y$$

$$\therefore x < q < y. \quad \square$$

Proof. $s := \sup A$, $t := \inf B$.

Step 1: $s \leq t$. $b \in B \implies b \in \mathbb{R}$ is an upper bound of $A \therefore s \leq b$ (s least). $\therefore s$ is a lower bound of $B \therefore s \leq t$ (t greatest).

Step 2: $t \leq s$. Suppose $s < t$. $\therefore s < q < t$ for some $q \in \mathbb{Q}$ (Lemma). $a \leq s < q$ for all $a \in A \therefore q$ is a rational upper bound of $A \therefore q \in B \therefore t \leq q$ (t lower bound of B).

$$t \leq q < t \quad (\text{contradiction})$$

$$\therefore t \leq s.$$

$$\text{Steps 1, 2} \implies \sup A = s = t = \inf B. \quad \square$$

Marking scheme (20 marks)

QRS suggested marking scheme.

- **3 marks:** Step 1 ($s \leq t$) complete.
- **3 marks:** Lemma correctly set up (choosing n with $1/n < y - x$, and M with $nx + M > 0$).
- **6 marks:** Lemma's construction of k via the least element of S , and the verification $k \leq nx < k + 1$.
- **3 marks:** correctly applying the Lemma in Step 2 (the rational q satisfies $s < q < t$, is an upper bound for A , and hence $q \in B$).
- **4 marks:** deriving the contradiction $t \leq q < t$ and concluding $t \leq s$.
- **1 mark:** the three writing expectations are met throughout.

Typical deductions: skipping the existence argument for the least element k_0 of S ; using a real (not necessarily rational) q in Step 2, which does not lie in B .